CIE-1

RV College of Engineering®
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Academic year 2023-2024 (Even Sem)

(OFFLINE CIE-I FOR II SEM ME STREAM)

DEPARTMENT OF CHEMISTRY

Date	13.05.24	Sem - I	CIE-I
Course Code	CM221IC	Maximum Test Marks	50
Course Name	CEM	Duration	90 Min
CHEMISTRY OF ENGINEERING MATERIALS (CEM)			

	Test Questions	M	BTL	CO
1	Illustrate the construction, charging, discharging and working mechanism of advanced battery technology, considering lithium cobalt oxide battery with neat labelled diagram.	7	2	1
2	Explain how photo catalyst water splitting contributes to the production of green hydrogen. Describe with neat labelled diagram the key components involved in this process and how they interact to facilitate the generation of hydrogen gas.	7	3	3
3	Imagine you are designing a new petrol engine for a high-performance sports car. Discuss the factors compression ratio and self-ignition in your analysis and design that contribute to engine knocking and propose innovative solutions to minimize or eliminate this phenomenon while maximizing engine power and efficiency.	7	4	1
4	Identify and highlight the significance of α-Methyl-Naphthalene and Iso-octane hydrocarbons in evaluation of knocking characteristics of petrol/diesel fuel.	7	3	3
5	What is gross calorific value and net calorific value? Derive an expression for Heat associated with steam and justify gross calorific value is greater than net calorific value.	7	2	2
6	Outline the principle, experimental procedure, model graph, electrodes and calculation in estimation of iron by potentiometric titration.	7	2	2
7	Suggest a volumetric titration method for the estimation of copper percentage in the Brass sample. Briefly outline the principle, reactions, procedure and calculation involved in the above method.	8	3	4

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

	Particulars	CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
Marks Distribution	Max Marks	14	14	14	08		21	21	08		

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Academic year 2023-2024 (Even Sem)

1RV23AS045

(OFFLINE CIE-II FOR II SEM MECHANICAL STREAM)

DEPARTMENT OF CHEMISTRY

Date	18.06.24	Sem - II	CIE-II
Course Code	CHY221IC	Maximum Test Marks	50
Course Name	**CEM**	Duration	90 Min
CHEMISTRY OF ENGINEERING MATERIALS (CEM)			

		Test Questions	M	BTL	CO
2	1	Illustrate the construction and working of working of EDLC super capacitor and pseudo capacitor along with materials used for such type of super capacitors.	7	3	1
3	2	Explain with example about the influence of corrosion product in the rate of corrosion? A piece of corroded steel plate was found in a submerged ocean vessel. It was estimated that the original area of the plate was 10 inch^2 and that approximately 2.3 kg had corroded away during submersion. Assuming a corrosion penetration rate of 180 mpy for this alloy in sea water, estimate the time of submersion in years. (given K in mpy = 534, density of steel is 7.9 g/cm^3)	7	3	3
3	3	With a neat labelled diagram propose the possible mechanism for rust formation using electrochemical theory. Interpret the resulting of rusting process with anodic and cathodic reactions.	7	2	1
3	4	Justify the following statements: a) Galvanization is preferred over Tinning for mechanical applications. b) For chrome plating, anode made up of chromium metal cannot be used. c) Pitting corrosion is localized and intense corrosion.	7	3	3
2	5	What are fuel cells? Outline the construction and working of direct methanol fuel cell along with its applications.	7	2	2
3	6	List out any two difference between electroplating and electroless plating. Explain the process with reactions of electroless plating of copper on a plastic object.	7	2	2
	7	Outline the principle, procedure, and instrumentation involved in the colorimetric determination of copper.	8	2	4

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars	CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Max Marks	08	21	14	7		29	21			

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CIE-3 (Improv)

Academic year 2023-2024 (Even Sem)

(OFFLINE IMPROVEMENT TEST FOR II SEM ME STREAM)

	Test Questions	M	BTL	CO
1	Describe catalytic and optical property in the context of nanomaterials and explain its importance in nano chemistry.	7	2	1
2	Illustrate the principle and methodology employed in synthesizing nanomaterials through solution combustion method and calculate the amount of the fuel needed to synthesize MgO nanomaterials from 5.5 g of $Mg(NO_3)_2.6H_2O$. Given: $Mg(NO_3)_2.6H_2O$ (Mol. Wt. =297.49) using glycine (NH_2CH_2COOH, Mol. Wt =75).	7	3	3
3	Justify the role of nitrogen in the synthesis of CNT. Explain the experimental setup along with mechanism for the synthesis of carbon nano tubes (CNT) from chemical vapor deposition.	7	2	1
4	In a petrol engine, the shock waves created due to knocking leads to the characteristic metallic sound. Interpret any two possible reasons which lead to knocking and suggest eco-friendly preventive measure?	7	3	3
5	Differentiate the following i) Gross Calorific value and Net calorific value of Fuels. ii) Octane number and Cetane number.	7	4	2
6	By applying the knowledge of chemistry of renewable energy sources, discuss the steps involved in the synthesis of biodiesel using transesterification process and mention any two disadvantages of biodiesel.	7	2	2
7	Discuss the principle, procedure and role of redox electrode before & after equivalence point for the estimation of iron in the given FAS solution using standard $K_2Cr_2O_7$. Plot the typical graphs and explain the chemistry behind the variation of potential.	8	3	4

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

	Particulars	CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
Marks Distribution	Max Marks	14	14	14	08		21	22	7		

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Academic year 2023-2024 (Even Sem)

(OFFLINE IMPROVEMENT TEST FOR II SEM ME STREAM)

DEPARTMENT OF CHEMISTRY

Date	1.07.24	Improvement Test + Quiz	Sem - II
Course Code	CM221IC	Maximum Test Marks	50 + 10
Course Name	CEM	Duration	120 Min
CHEMISTRY OF ENGINEERING MATERIALS (CEM)			

	Quiz Questions	**M**	**BTL**	**CO**
1	Power alcohol reduces knocking in petrol IC engine, justify.	1	3	1
2	Mention any one catalyst used in the hydrogen production by photocatalytic water splitting.	1	2	4
3	Gross calorific value is greater than net calorific value, justify.	1	3	3
4	Differentiate between rocket propellant and explosives.	1	4	4
5	Mention any one disadvantage of biodiesel.	1	2	2
6	In the Hummer's method used for graphene oxide preparation, identify the role of reagent $KMnO_4$.	1	3	3
7	What is aerogel?	1	2	2
8	Write the different types of single walled carbon nanotubes.	1	2	2
9	Identify the hybridization state of carbon in CNT's after functionalization with $-NH_2$ group.	1	3	3
10	List any one of the applications of nano lubricant.	1	2	1

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars	CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Max Marks	2	3	3	2		5	4	1		

P.T.O

RV... (An A...) II Semester B. E...

CM221IC

USN | 1 | RV | 23 | AS | 045 |

RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

II Semester B. E. Regular / Supplementary Examinations Aug-2024

COMMON TO AS /CH / IM / ME

CHEMISTRY OF ENGINEERING MATERIALS

Time: 03 Hours *Maximum Marks: 100*

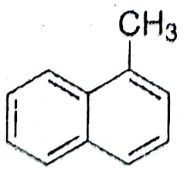
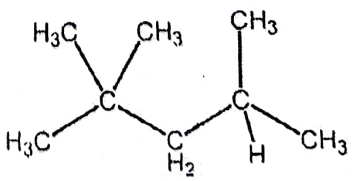
Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 & 11 are compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10, and 11 lab components (compulsory).

PART-A

			M	BT	CO
1	1.1	Why *HCV* is always higher than *LCV*?	01	3	3
	1.2	Out of the two additives ethyl nitrate and ethanol identify which can improve the cetane number.	01	3	2
	1.3	Identify the battery characteristic used to predict the life span of secondary battery.	01	2	3
	1.4	Write the anodic reaction of aqueous lithium air battery.	01	3	2
	1.5	Identify a metal from the given list which is protected by a layer of its own oxide. List of metals are *Cu, Au, Fe* and *Al*	01	3	1
	1.6	Justify why Iron does not rust when exposed to dry air.	01	3	3
	1.7	After the functionalization identify the hybridization of carbon atom at the functionalized site.	01	2	3
	1.8	Justify gold nanoparticles exhibit different colors.	01	4	3
	1.9	Give reason why glass transition temperature of polypropylene is lower compared to polyvinylchloride.	01	2	2
	1.10	What is the commercial name of Phenol-formaldehyde resin?	01	2	1

PART B

2	a	What is knocking in petrol IC engine and diesel engine? By applying the knowledge of chemistry, Identify and highlight the significance of the below hydrocarbon in evaluation of knocking characteristics of petrol/diesel fuel. Mention one adverse of knocking.   *α*-Methyl-Naphthalene Iso-octane	07	2	2
	b	What is the significance of green hydrogen? Discuss the production and storage of green hydrogen.	07	2	2

3	a	Outline the construction and working of the rechargeable $LiCoO_2$ battery with suitable diagram. With the help of reversible redox reactions, discuss the $Li+$ ion exchange mechanism between anode and cathode.	07	2	2
	b	With neatly labeled diagram explain Methanol-Oxygen fuel cell and the mechanism of conversion of chemical energy into electrical energy using anodic and cathodic reactions taking place in the presence of H_2SO_4 electrolyte. Discuss any one application of fuel cell.	07	2	2
		OR			
4	a	Outline the fabrication and working of the lithium-air battery with a suitable diagram and reactions. Discuss its limitation.	07	2	2
	b	With neat labeled diagram explain the construction and working of Quantum dot sensitized solar cell.	07	2	2
5	a	How rate of corrosion is affected by the following factors i) Corrosion product ii) Relative anodic and cathodic area iii) Temperature	07	3	3
	b	What are the advantages of electroless plating over electroplating? Explain the process of electroplating of chromium for engineering applications and indicate the reasons for not employing chromium as anode.	07	2	2
		OR			
6	a	What is anodic metallic coating? Justify the need of galvanized metal sheets used for outdoor purpose. Explain the process of galvanization.	07	3	3
	b	Discuss with neat labeled diagram the corrosion of iron rod exposed in environment based on electrochemical theory.	07	2	2
7	a	Discuss the reactions and steps involved with neat labeled diagram in the preparation of nano materials by sol-gel method.	07	2	1
	b	Illustrate the synthesis of grapheme by Hummer's method. Comment on its mechanical and optical properties. Mention any one application of grapheme.	07	2	1
		OR			
8	a	Discuss the principle and procedure in the preparation of nano materials by solution combustion synthesis.	07	2	1
	b	Illustrate the synthesis of carbon nanotube by modified CVD method with neat labeled diagram. Mention any two application of carbon nanotube.	07	2	1
9	a	What is biodegradable polymer? Explain the synthesis of poly lactic acid with application.	07	2	2
	b	Discuss the synthesis of glass fiber. How it can be used in polymer composites and list two applications.	07	2	2
		OR			

10	a	What is glass transition temperature of a polymer? How the following factors affect glass transition of polymer i) Chemical Structure ii) Stereo regularity iii) Flexibility	07	2	2
	b	Discuss the synthesis of carbon fiber. How it can be used in polymer composites and list two applications?	07	2	2
		LAB COMPONENT			
11	a	Discuss the procedure for conductometric titration of strong acid with strong base procedure. Plot the model graph and explain the nature of graph.	10	2	3
	b	Explain the colorimetry principle, procedure and calculation involved in the estimation of copper from copper sulphate solution.	10	2	3

